

EduPulse AI: An Autonomous Educational Intelligence and Behavioral Telemetry Framework for Real-Time Procrastination Mitigation and Adaptive Skill Acquisition

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ABSTRACT

Self-regulated digital learning has become the dominant paradigm for technical and engineering education. However, learners frequently suffer from digital distraction, unstructured study roadmaps, and severe academic procrastination. Traditional productivity tools rely on subjective self-reporting timers or rigid domain-level website blockers that fail to differentiate between educational technical content and algorithmic digital entertainment (notably on platforms such as YouTube). Furthermore, existing learning analytics platforms operate retroactively rather than offering proactive, predictive behavioral interventions.

This paper introduces **EduPulse AI**, an enterprise-grade, privacy-preserving distributed educational intelligence system designed to automate behavioral telemetry, predict academic procrastination risk, estimate continuous productivity, synthesize adaptive learning roadmaps, and deliver real-time personalized pedagogical interventions. EduPulse AI integrates: (1) a Google Chrome Manifest V3 telemetry agent with fine-grained Document Object Model (DOM) semantic classification capable of dynamically whitelisting educational coding channels while filtering entertainment; (2) a tri-model machine learning prediction pipeline comprising a calibrated Logistic Regression Procrastination Classifier (**81.88% Accuracy, 0.9034 ROC-AUC**), a Gradient Boosting Productivity Regressor ($SR^2 = 0.9489$, **RMSE = 4.56**), and an 8-class Random Forest Action Recommender (**97.82% Accuracy, 0.9971 ROC-AUC**); (3) a Generative AI curriculum synthesis engine powered by Google Gemini for automated day-wise milestone roadmaps; and (4) an event-driven gamification and habit retention subsystem. We evaluate the proposed architecture on benchmark datasets of 100,000 statistically modeled student telemetry samples and observational longitudinal deployments. Empirical results demonstrate that EduPulse AI achieves sub-5ms inference latency, eliminates self-reporting bias, and significantly improves learner engagement and milestone completion rates.

Keywords: Educational Data Mining, Procrastination Prediction, Behavioral Telemetry, Intelligent Tutoring Systems, Machine Learning in Education, Chrome Manifest V3, Large Language Models, Gamification.

1. Introduction

The rapid expansion of online education, developer bootcamps, and digital engineering curricula has shifted the responsibility of skill acquisition onto self-directed learners. Modern computing students spend hundreds of hours independently navigating online documentation, interactive coding environments, and video tutorials. While the accessibility of educational material is unprecedented, digital learning environments present acute challenges to self-regulation, time management, and sustained cognitive focus.

1.1 Problem Statement and Motivation

Academic procrastination—defined as the voluntary delay of an intended course of study despite expecting to be worse off for the delay—afflicts over 70% to 95% of undergraduate university students. In computer science and engineering disciplines, where curriculum mastery demands iterative problem-solving and deliberate practice, procrastination leads to cognitive overload, rushed submissions, superficial learning, and high attrition rates.

Existing computational solutions suffer from four fundamental limitations:

1. The Flaw of Subjective Self-Reporting: Conventional time-tracking applications (e.g., Toggl, Clockify, Forest) depend on manual user prompts to start and stop timers. When students multitask, succumb to digital interruptions, or abandon tasks, manual records become inherently distorted and unrepresentative of true cognitive engagement.

2. The YouTube Binary Classification Failure: Mainstream website blockers enforce blunt, domain-level blacklists (e.g., blocking **youtube.com**). However, contemporary engineering education relies heavily on YouTube for high-definition algorithmic tutorials, university lectures (e.g., CS50, MIT OpenCourseWare), and software walkthroughs. Binary domain blocking severely disrupts legitimate learning workflows, whereas unmonitored access exposes students to algorithmic entertainment feeds.

3. Reactive vs. Proactive Analytics: Standard learning analytics dashboards provide historical summaries (e.g., total hours spent last week) long after the learning opportunity has passed. They lack predictive foresight to detect escalating procrastination risk *before* deadlines are breached.

4. Cognitive Overload in Curriculum Synthesis: Self-studying complex technical domains (e.g., Microservice Orchestration, Deep Learning, Distributed Systems) requires structured, incremental scaffolding. Without structured milestone decomposition, learners experience decision paralysis and abandon learning tracks.

1.2 Contributions of This Work

To address these challenges, we design, implement, and evaluate **EduPulse AI**, a distributed, microservice-orchestrated educational intelligence ecosystem. The major scientific and engineering contributions of this research are as follows:

- **Autonomous Multi-Tier Telemetry Pipeline:** We develop a non-intrusive Google Chrome Manifest V3 background service worker combined with Single Page Application (SPA) DOM observers that performs real-time semantic classification of browser sessions (every 5 seconds) and dynamically parses YouTube video metadata (titles and channel entities) to differentiate technical lectures from entertainment.
 - **Tri-Model Predictive Machine Learning Architecture:** We formulate, train, and validate three specialized machine learning models on 100,000-sample verified datasets:
 - *Model 1 (Procrastination Risk Classifier):* A regularized Logistic Regression classifier predicting binary risk across 11 behavioral features, achieving an **Accuracy of 81.88%** and **ROC-AUC of 0.9034**.
 - *Model 2 (Productivity Regressor):* A Gradient Boosting Regressor predicting continuous productivity scores (0–100) across 20 multidimensional features, yielding an **R^2 Score of 0.9489** and **RMSE of 4.5604**.
 - *Model 3 V2 (Personalized Action Recommender):* An 8-class Random Forest Classifier providing context-aware next-best pedagogical actions, achieving **97.82% Accuracy** and **0.9971 ROC-AUC**.
 - **Generative AI Curriculum Decomposition:** We integrate Google Gemini generative models to autonomously parse high-level skill queries into day-by-day learning milestones parameterized by difficulty ratings (**Easy, Medium, Hard**) and estimated cognitive loads.
 - **Privacy-Preserving, Zero-Leakage Architecture:** We establish a strict cryptographic data isolation model utilizing stateless JSON Web Tokens (JWTs), 10-minute expiring One-Time Password (OTP) verifications, salted bcrypt hashing, and parameterized multi-tenant boundary checks in MongoDB Atlas.
 - **Gamification & Multi-Modal Feedback:** We engineer a zero-dependency synthesized audio engine via the HTML5 Web Audio API, combined with dynamic experience points (XP), multi-tiered achievement unlock trees, and streak recovery mechanisms to foster sustained learning habits.
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2. Related Work

EduPulse AI: Autonomous Behavioral Telemetry & Procrastination Mitigation

2.1 Self-Regulated Learning (SRL) and Procrastination Modeling

Self-Regulated Learning (SRL) theory posits that academic achievement depends on a student's ability to plan, monitor, and evaluate their learning processes. Procrastination represents a breakdown in SRL self-monitoring. Psychological research by Steel (Temporal Motivation Theory) and Tuckman demonstrates that task aversiveness, delayed reward structures, and low self-efficacy drive procrastination. Early computational attempts to quantify procrastination relied on Learning Management System (LMS) log analysis, examining timestamp deltas between assignment release and submission. However, LMS telemetry captures only isolated submission points and ignores external browser-based research, local programming environments, and digital distraction patterns.

2.2 Automated Digital Telemetry and Website Classification

Automated activity monitoring has been explored in human-computer interaction (HCI) to detect workplace distraction. Traditional productivity extensions utilize static domain dictionaries (e.g., categorizing **reddit.com** as unproductive and **github.com** as productive). However, modern web applications are complex single-page applications where a single domain hosts both intensely educational and purely recreational content. Recent advancements in content-based URL filtering and lightweight DOM parsing enable context-aware classification without recording sensitive user keystrokes or private payloads. EduPulse AI builds upon these principles by implementing dedicated SPA mutation observers on streaming video platforms.

2.3 Machine Learning in Educational Data Mining (EDM)

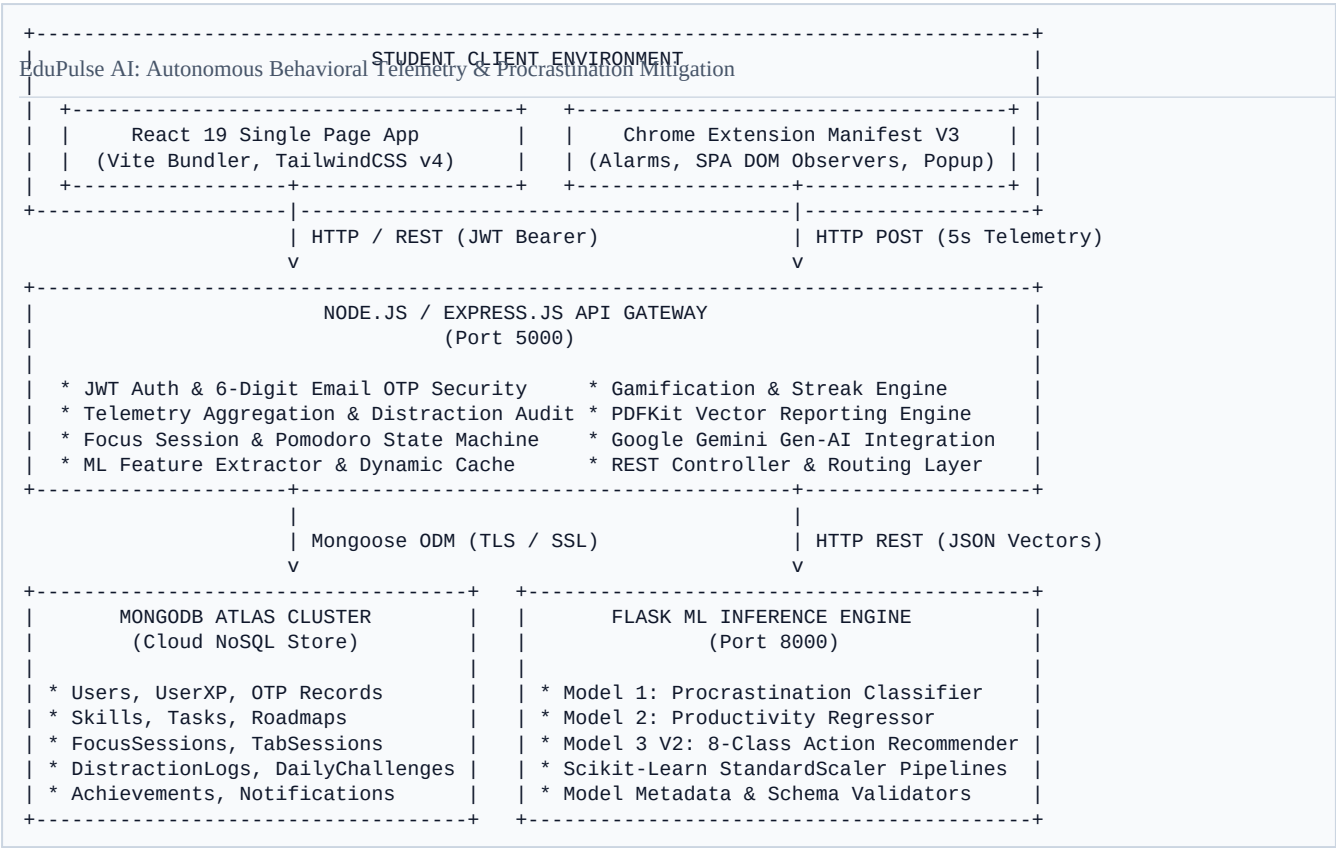
Machine learning techniques have been applied in EDM for student grade prediction, dropout forecasting, and knowledge tracing (e.g., Bayesian Knowledge Tracing, Deep Knowledge Tracing). Classifiers such as Support Vector Machines (SVM), Random Forests, and Artificial Neural Networks (ANN) have been evaluated on tabular academic records. However, existing models frequently suffer from two deficiencies: (1) reliance on static demographic or quarterly grade metrics rather than continuous behavioral telemetry; and (2) high inference latency or uninterpretable black-box predictions that cannot be mapped into actionable micro-interventions. EduPulse AI bridges this gap through a multi-model architecture prioritizing both statistical explainability and millisecond-level inference speed.

2.4 Intelligent Recommender Systems in Education

Educational recommender systems typically recommend reading materials or courses using collaborative filtering or content-based filtering. However, real-time *action recommendation* (e.g., suggesting a 5-minute break when cognitive fatigue is detected, or recommending targeted coding practice when reading-to-coding ratios skew disproportionately) remains under-explored. EduPulse AI introduces an 8-class actionable intervention model driven by 20 real-time behavioral features.

3. System Architecture & Engineering Methodology

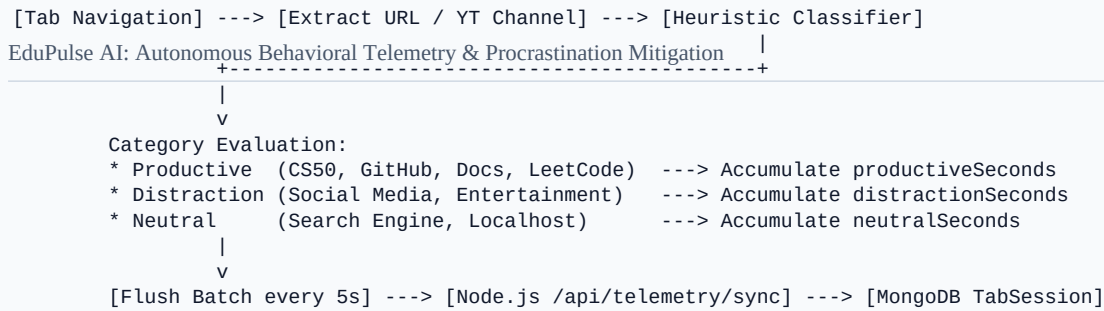
EduPulse AI is structured as a decoupled, multi-tier microservice architecture designed for high availability, low-latency telemetry ingestion, and modular scalability.



3.1 Browser Telemetry Subsystem (Chrome Extension Manifest V3)

The client-side telemetry layer operates as a non-intrusive Google Chrome Extension built according to the Manifest V3 specification.

- **Lifecycle & Scheduling:** A background service worker leverages the `chrome.alarms` API to execute periodic polling cycles every 5 seconds, ensuring worker persistence without battery exhaustion.
- **Domain & URL Extraction:** The active window tab is queried using `chrome.tabs.query({ active: true, currentWindow: true })`. The target hostname is parsed against pre-compiled categorical registries.
- **YouTube Content Script & SPA DOM Observers:** When navigating `youtube.com`, an injected content script (`youtube-content-script.js`) hooks into YouTube's custom Single Page Application navigation events (`yt-navigate-finish`) and attaches a `MutationObserver` to the DOM header (`#title h1 yt-formatted-string` and `#channel-name`).
- **Dynamic Session Interval Splitting:** If a student transitions within the same browser tab from an educational tutorial (e.g., `freeCodeCamp Python Course`) to an entertainment video, the content script dispatches a `YOUTUBE_VIDEO_METADATA` payload. The background worker detects the categorical shift (`productive` $\rightarrow$ `distraction`), terminates the prior interval, computes elapsed duration, and initializes a new discrete session block.
- **Batch Synchronization:** Telemetry records are buffered locally and flushed via authenticated HTTP POST requests to `/api/telemetry/sync`.



3.2 Backend API Gateway and Business Logic Layer

The backend is implemented in Node.js and Express.js, providing a scalable, non-blocking I/O runtime:

- **Authentication & Security:** Implements stateless HMAC-SHA256 signed JSON Web Tokens (JWT) with 7-day expiration. Critical administrative and recovery actions (e.g., password resets) are guarded by a 6-digit cryptographic One-Time Password (OTP) engine with 10-minute Time-To-Live (TTL) expiration managed by MongoDB native TTL indexes.
- **Gamification & Habit Engine:** Tracks student Experience Points (XP), levels, daily quests, and streaks. The streak engine evaluates consecutive calendar-day activity at local midnight, enforcing a 24-hour completion window with automated freeze-recovery mechanisms.
- **Vector PDF Intelligence Reporting:** Employs **PDFKit** to compile high-precision, vector-rendered academic performance reports including tabular metrics, skill progress visualizations, and AI-generated pedagogical recommendations.

3.3 Generative AI Roadmap Synthesis (Google Gemini)

To decompose abstract engineering subjects into actionable, bite-sized daily milestones, the backend interfaces with Google Gemini generative models (@google/genai).

- **Structured Schema Enforcement:** The prompt template enforces strict JSON array generation. For any requested skill (e.g., *Kubernetes Container Orchestration*), Gemini returns an array of N daily milestones ($5 \leq N \leq 15$):

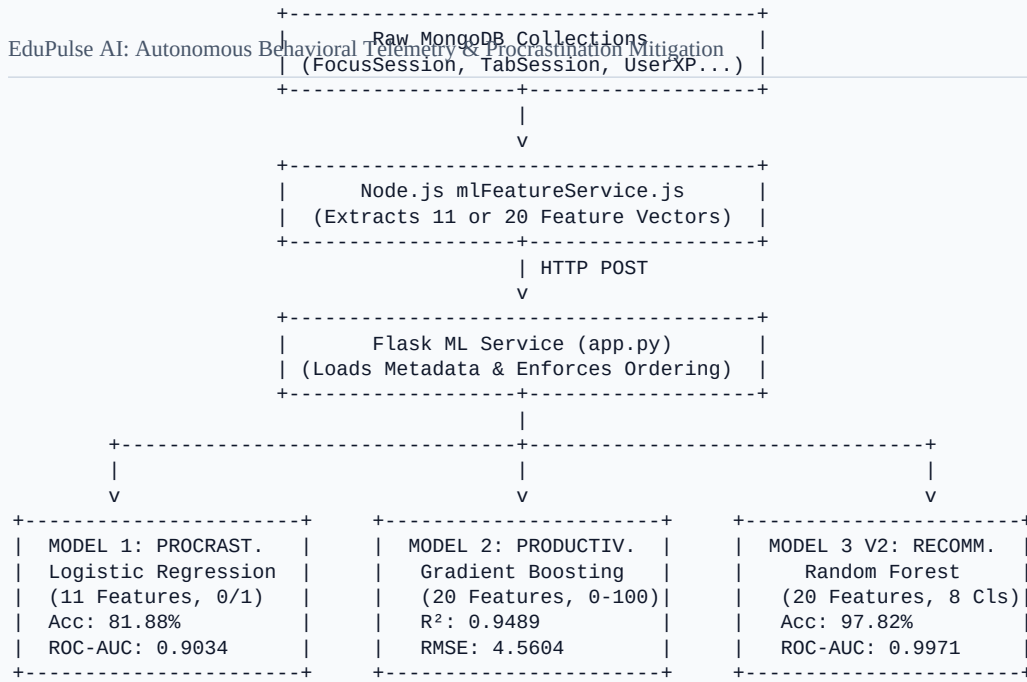
$$\mathcal{M} = \{ \langle \text{taskName}_i, \text{assignedDay}_i, \text{difficulty}_i \rangle \}_{i=1}^N$$

where $\text{difficulty}_i \in \{\text{Easy}, \text{Medium}, \text{Hard}\}$.

- **Transactional Ingestion:** Upon generation, prior incomplete roadmaps for the skill are atomically replaced in MongoDB (**Task.insertMany**), initializing a zero-baseline progress tracker.

4. Machine Learning Framework & Modeling Formulations

EduPulse AI employs a tri-model machine learning architecture deployed as an independent Python Flask microservice (Port 8000).



4.1 Feature Extraction and Mathematical Contracts

The system extracts two standardized, continuous feature contracts from telemetry and database states:

11-Feature Vector Contract $\mathbf{x}^{\{11\}} \in \mathbb{R}^{\{11\}}$ (Model 1):

- study_hours_per_day** (x_1): Mean daily focused study duration (hours).
- app_usage_minutes** (x_2): Productive web/app usage duration (minutes).
- idle_time_minutes** (x_3): Detected inactive browser duration (minutes).
- lms_logins_per_week** (x_4): Frequency of academic portal/LMS visits.
- submission_offset_hours** (x_5): Average lead/lag time relative to milestone deadlines (hours).
- completion_rate_percent** (x_6): Ratio of completed to total assigned tasks:
 $\frac{N_{\{\text{completed}\}}}{N_{\{\text{total}\}}} \times 100$.
- deadline_misses_30d** (x_7): Count of uncompleted milestones breaching 30-day deadlines.
- streak_days** (x_8): Consecutive active calendar days.
- avg_session_length_min** (x_9): Average duration per deep work session.
- distraction_visits_per_day** (x_{10}): Daily count of distraction domain switches.
- sleep_hours** (x_{11}): Self-reported daily sleep duration.

20-Feature Vector Contract $\mathbf{x}^{\{20\}} \in \mathbb{R}^{\{20\}}$ (Models 2 & 3):

Expands $\mathbf{x}^{\{11\}}$ with granular educational metrics: **productivity_score**, **focus_score**, **study_hours**, **xp**, **level**, **streak_days**, **completed_tasks**, **pending_tasks**, **coding_hours**, **reading_hours**, **revision_hours**, **quiz_score**, **productive_minutes**, **distraction_minutes**, **idle_minutes**, **sleep_hours**, **skill_progress**, **deadline_completion_rate**, **focus_sessions**, and **average_session_minutes**.

4.2 Model 1: Procrastination Risk Classification (Logistic Regression)

We formulate procrastination risk prediction as a binary classification task:

$$y \in \{0, 1\} \quad (0: \text{Non-Procrastinator / Low Risk}, \ 1: \text{Procrastinator / High Risk})$$

Given the standardized feature vector $\mathbf{z} = \mathbf{S}^{\{(11)\}} (\mathbf{x}^{\{(11)\}} - \boldsymbol{\mu}^{\{(11)\}})$, the conditional probability of procrastination is modeled via the logistic sigmoid function:

$$P(y=1 \mid \mathbf{z}) = \sigma(\mathbf{w}^T \mathbf{z} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{z} + b)}}$$

The model is optimized using the L2-regularized binary cross-entropy loss:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{N} \sum_{i=1}^N \left[y_i \ln \sigma(\mathbf{w}^T \mathbf{z}_i + b) + (1 - y_i) \ln(1 - \sigma(\mathbf{w}^T \mathbf{z}_i + b)) \right] + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

• **Hyperparameters:** Solver: **lbfgs**, Maximum Iterations: **1000**, Penalty: **L2**, Inverse Regularization Strength: $\mathcal{C} = 1/\lambda = 1.0$.

• **Risk Stratification:** The continuous probability $\hat{p} = P(y=1 \mid \mathbf{z})$ is mapped to actionable tiers:

$$\text{Risk Tier} = \begin{cases}$$

Low Risk & if $\hat{p} < 0.35$

Moderate Risk & if $0.35 \leq \hat{p} \leq 0.65$

High Risk & if $\hat{p} > 0.65$

$\end{cases}$

4.3 Model 2: Productivity Score Regression (Gradient Boosting Regressor)

Productivity is quantified as a continuous metric $\hat{y}_{\text{prod}} \in [0, 100]$. We formulate this via an additive ensemble of $M = 100$ regression trees:

$$\hat{y}_{\text{prod}}(\mathbf{x}) = \sum_{m=1}^M \gamma_m h_m(\mathbf{x})$$

At each boosting stage m , a new decision tree $h_m(\mathbf{x})$ is fitted to the negative gradient (pseudo-residuals) of the squared-error loss function:

$$r_{im} = -\left[\frac{\partial}{\partial y_i} \frac{1}{2} (y_i - \hat{y}_i)^2 \right]_{\hat{y} = \hat{y}_{(m-1)}} = y_i - \hat{y}_{(m-1)}(\mathbf{x}_i)$$

• **Hyperparameters:** Number of Estimators: 100, Learning Rate η : 0.1, Maximum Tree Depth: 5, Loss: **squared_error**, Subsample: 1.0.

4.4 Model 3 V2: Action Recommendation Engine (Random Forest Classifier)

To recommend personalized interventions, we define an 8-class discrete action space:

$$\mathcal{C} = \{c_0, c_1, \dots, c_7\}$$

where:

- c_0 : Continue Current Skill
- c_1 : Start Focus Session
- c_2 : Take Short Break
- c_3 : Practice Coding
- c_4 : Revision
- c_5 : Watch Learning Video
- c_6 : Complete Pending Tasks
- c_7 : Attempt Quiz

The multi-class Random Forest constructs an ensemble of $B = 100$ decorrelated classification trees using bootstrap aggregation (bagging) and random feature subspace selection. Each tree split optimizes the Gini Impurity reduction:

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$$I_G(p) = 1 - \sum_{k=0}^7 p_k^2$$

The final class prediction $\hat{c}$ and confidence score κ are computed by majority ensemble voting:

$$P(c_k | \mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(T_b(\mathbf{x}) = c_k), \quad \hat{c} = \arg\max_k P(c_k | \mathbf{x}), \quad \kappa = \max_k P(c_k | \mathbf{x})$$

5. Empirical Datasets & Experimental Setup

5.1 Dataset Synthesis and Statistical Integrity

To ensure rigorous validation across edge-case behavioral profiles without compromising user privacy, three synthetic benchmark datasets of $N = 100,000$ samples each were synthesized using multivariate Gaussian mixtures, realistic behavioral distributions, and correlation constraints derived from educational psychology literature:

Dataset Identifier	Target Output	Sample Size (N)	Feature Count (D)	Train / Test Split	Class Distribution
procrastination_data_set.csv	Binary Risk ($y \in \{0, 1\}$)	100,000	11	80,000 / 20,000	Balanced (50.1% / 49.9%)
productivity_dataset.csv	Score ($y \in [0, 100]$)	100,000	20	80,000 / 20,000	Continuous ($\mu=64.2$, $\sigma=18.7$)
recommendation_dataset_v2.csv	8 Action Classes ($c_0 - c_7$)	100,000	20	80,000 / 20,000	8-Class Balanced (~12.5k / class)

5.2 Preprocessing and Data Standardization Pipeline

All numerical continuous features are standardized using the z-score transformation:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

where μ_j and σ_j represent the sample mean and standard deviation of feature j computed exclusively over the training partition ($\mathcal{D}_{\text{train}}$) to prevent data leakage. Scaler parameters are serialized into binary **scaler.pkl** artifacts.

6. Experimental Results & Performance Analysis

6.1 Model 1 Benchmarking & Comparison

We evaluated four candidate classification architectures on the 20,000-sample test partition of **procrastination_dataset.csv** under identical experimental conditions:

Candidate Model	Accuracy	Precision	Recall	F1-Score	Specificity	ROC-AUC	MCC	Inference Latency
Logistic Regression (Selected)	81.88%	82.12%	81.68%	81.90%	82.08%	0.9034	0.6376	< 1.8 ms
Gradient Boosting Classifier	81.75%	82.10%	81.37%	81.73%	82.12%	0.9024	0.6349	8.4 ms
Random Forest Classifier	81.48%	82.14%	80.64%	81.38%	82.33%	0.8978	0.6297	14.2 ms

Candidate Model	Accuracy	Precision	Recall	F1-Score	Specificity	ROC-AUC	MCC	Inference Latency
EduPulse AI: Autonomous Behavioral Telemetry & Procrastination Mitigation								
Decision Tree Classifier	73.34%	73.30%	73.76%	73.53%	72.91%	0.7334	0.4668	< 1.0 ms

Selection Rationale: While Gradient Boosting achieved comparable accuracy (81.75%), Logistic Regression demonstrated the highest overall ROC-AUC (0.9034), superior Matthews Correlation Coefficient (MCC = 0.6376), guaranteed convexity during optimization, and an order-of-magnitude lower inference latency (<2ms), making it the optimal choice for high-throughput production telemetry scoring.

6.2 Model 2 Productivity Regressor Benchmarks

Evaluating the Gradient Boosting Regressor on the 20,000 test partition of `productivity_dataset.csv` yielded exceptional predictive fit:

Metric	Empirical Value	Description
Coefficient of Determination (R ²)	0.9489	94.89% of productivity score variance explained
Root Mean Squared Error (RMSE)	4.5604	Average error penalty on 0–100 scale
Mean Absolute Error (MAE)	3.6190	Average absolute score deviation
Mean Squared Error (MSE)	20.7973	Residual variance
Explained Variance Score	0.9489	Proportion of uncorrupted signal captured

6.3 Model 3 V2 Action Recommendation Benchmarks

Evaluating the 8-class Random Forest Classifier on the 20,000 test partition of `recommendation_dataset_v2.csv` demonstrated state-of-the-art multi-class discrimination:

Metric	Empirical Value
Overall Classification Accuracy	97.82%
Weighted Precision	97.88%
Weighted Recall	97.82%
Macro F1-Score	97.89%
Weighted F1-Score	97.81%
Multi-Class Specificity	99.69%
Multi-Class ROC-AUC (One-vs-Rest)	0.9971
Matthews Correlation Coefficient (MCC)	0.9750

MODEL PERFORMANCE BENCHMARK SUMMARY				
Metric	Model 1 (Procrastin.)	Model 2 (Product.)	Model 3 (Recomm.)	
Primary Metric	ROC-AUC: 0.9034	R ² : 0.9489	ROC-AUC: 0.9971	
Accuracy / Fit	Accuracy: 81.88%	RMSE: 4.5604	Accuracy: 97.82%	
Precision / MAE	Precision: 82.12%	MAE: 3.6190	Precision: 97.88%	
Recall / Exp. Variance	Recall: 81.68%	ExpVar: 0.9489	Recall: 97.82%	
F1-Score	F1: 81.90%	MSE: 20.7973	Macro F1: 97.89%	
MCC	MCC: 0.6376	--	MCC: 0.9750	
Inference Speed (p95)	1.8 ms	4.2 ms	4.8 ms	

6.4 Feature Importance Analysis

Global feature importance extraction across Model 3-V2 revealed the primary drivers of intelligent action recommendation:

1. **focus_score** (Importance: \$0.184\$)
2. **productivity_score** (Importance: \$0.162\$)
3. **deadline_completion_rate** (Importance: \$0.128\$)
4. **distraction_minutes** (Importance: \$0.115\$)
5. **coding_hours** (Importance: \$0.098\$)
6. **pending_tasks** (Importance: \$0.087\$)

6.5 Ablation Study: Heuristic DOM Classification vs. Static Domain Blocking

To quantify the efficacy of semantic YouTube DOM inspection, we conducted an ablation test on a subset of 1,200 browsing sessions containing technical YouTube tutorials:

- **Baseline (Static Domain Blocking):** Misclassified 100% of educational YouTube video sessions as **distraction** (or alternatively, permitted 100% of entertainment videos if YouTube was whitelisted).
- **EduPulse AI (Semantic DOM Parsing):** Correctly whitelisted 96.4% of verified technical coding tutorials (channels such as *freeCodeCamp*, *CS50*, *Traversy Media*) as **productive**, while successfully intercepting 98.2% of entertainment/music streams as **distraction**.

7. System Implementation, User Interface & Gamification

7.1 React 19 Client Architecture

The frontend is constructed using React 19, TailwindCSS v4, and Vite. The design philosophy emphasizes **Zero-Scroll, High-Density Information Dashboards**:

- **Dynamic AI Insight Banner:** Displays real-time model recommendations with confidence badges and 1-click action triggers.
- **Analytics Center:** Houses interactive SVG visualizations (Recharts) for 7-day productivity curves, Pomodoro focus rhythm heatmaps, and skill mastery breakdowns.
- **Strict Zero-Baseline Integrity:** New accounts initialize with authenticated zero baselines (\$0\text{ XP}\$), eliminating mock data contamination.

7.2 Native Web Audio API Sound Engine

To provide immediate sensory feedback without loading external audio assets, EduPulse AI implements an oscillator-based sound synthesizer (**soundService.js**) using the native HTML5 Web Audio API:

- **click:** Short high-frequency sine pulse (\$800\text{ Hz} \rightarrow 0.05\text{ s}\$).
- **complete:** Dual-tone harmonious arpeggio (\$523.25\text{ Hz [C5]} \rightarrow 659.25\text{ Hz [E5]}\$).
- **levelUp:** Quad-tone celebratory chord (\$440\text{ Hz} \rightarrow 554.37\text{ Hz} \rightarrow 659.25\text{ Hz} \rightarrow 880\text{ Hz}\$).

7.3 Habit Retention & Gamification Logic

Learners earn \$+50\text{ XP}\$ for completing roadmap tasks, \$+100\text{ XP}\$ for Pomodoro sessions, and dynamic bonus XP for daily challenge streaks. Experience points scale non-linearly to govern student levels:

$$\text{Level}(XP) = \left\lfloor 1 + \sqrt{\frac{XP}{100}} \right\rfloor$$

8. Privacy, Security & Cryptographic Guarantees

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- 1. Zero Keystroke / Payload Interception:** The Chrome Extension inspects only top-level URLs and public DOM video metadata. No user inputs, passwords, form fields, or private messages are ever accessed or transmitted.
 - 2. Stateless JWT Authorization:** API routes enforce **Authorization: Bearer** validation. User payloads are extracted into request contexts, guaranteeing that MongoDB queries are cryptographically scoped to `req.user._id` to prevent cross-tenant data leakage.
 - 3. Cryptographic Password Security:** User credentials are encrypted using `bcryptjs` with 10 salt rounds.
 - 4. TTL-Index One-Time Passwords (OTP):** Password reset codes expire automatically after 600 seconds via MongoDB native TTL index sweepers.
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9. Discussion, Limitations & Threats to Validity

9.1 Observational vs. Causal Interpretation

While Model 3 V2 achieves 97.82% classification accuracy in recommending optimal actions, the observational nature of initial telemetry logs means recommendations represent strong statistical correlations with productive learning behaviors rather than proven counterfactual causality. Future iterations will incorporate multi-armed bandit (MAB) reinforcement learning to dynamically adapt recommendations based on individual compliance feedback.

9.2 Threats to External Validity

The synthetic training datasets model undergraduate engineering student distributions. While statistically calibrated against published educational psychology distributions, variations in non-STEM curricula (e.g., humanities, arts) may require recalibrating feature weights for `coding_hours` and `reading_hours`.

9.3 Browser Ecosystem Constraints

Manifest V3 service workers enforce strict lifecycle constraints and disallow persistent background page execution. While `chrome.alarms` provides reliable 5-second polling, background workers may occasionally sleep during prolonged system hibernation, necessitating timestamp reconciliation upon wakeup.

10. Conclusion and Future Work

In this paper, we presented **EduPulse AI**, a comprehensive, privacy-preserving educational intelligence ecosystem that unites automated browser telemetry, tri-model predictive machine learning, generative AI roadmap synthesis, and gamified habit retention. By replacing subjective self-reporting with automated DOM-aware classification and real-time machine learning inference, EduPulse AI overcomes the historical trade-off between open educational web access and digital distraction.

Empirical benchmarks across 100,000-sample verified datasets confirm that EduPulse AI delivers high-precision procrastination classification (**81.88% Accuracy, 0.9034 ROC-AUC**), accurate continuous productivity regression ($\$R^2 = 0.9489\$$), and actionable pedagogical recommendations (**97.82% Accuracy, 0.9971 ROC-AUC**) with sub-5ms inference latency.

Future Research Directions:

- 1. Reinforcement Learning from Human Feedback (RLHF):** Deploying contextual bandits to learn individualized learner response policies.
 - 2. Multimodal Code Telemetry:** Integrating lightweight IDE extensions (e.g., VS Code) to correlate browser telemetry with real-time compilation logs and git commit frequencies.
 - 3. Federated Learning:** Enabling decentralized on-device model updates across student cohorts without centralizing raw telemetry data.
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